To be able to do so, you need to add the following to your infrastructure:

a) A file server with a suitable protocol (e.g., Samba, NFS, etc)
b) A client-side software capable of mounting the shared file system, locally

For the purpose of this article, we will use NFS as the protocol for file sharing since both the server and the client-side software are available in Linux Operating System. Let's create a share on the LDAP server itself for the users to keep their data.

To prepare a basic NFS share on the NFS server, you need to go through the following steps:

a) Decide on the folder to be shared
b) Export it

> **Note:** We will not go into NFS details; our aim is to present roaming profiles to users

To create a central file share on the NFS server, let us create a new folder called myusersdata. The following commands on the NFS server should achieve this:

```
[root@vbg ~]# mkdir /myusersdata

[root@vbg ~]# echo "/myusersdata   *(rw)" >> /etc/exports

[root@vbg ~]# service nfs restart
Shutting down NFS mountd:                          [FAILED]
Shutting down NFS daemon:                          [FAILED]
Shutting down NFS quotas:                          [FAILED]
Shutting down NFS services:                        [FAILED]
Starting NFS services:                             [  OK  ]
Starting NFS quotas:                               [  OK  ]
Starting NFS daemon:                               [  OK  ]
Starting NFS mountd:                               [  OK  ]

[root@vbg ~]# showmount -e localhost
Export list for localhost:
/myusersdata *
```

Now we need to create home directories for all the users created in LDAP; an easy way to do this would be to check the valid users from the client.

On the client, let's take a look at the last four users:

```
[root@rdso ~]# getent passwd | tail -n 4
vbg1:x:501:501:vbg1:/home/vbg1:/bin/bash
websms:x:502:502:websms:/home/websms:/bin/bash
rajveer:x:503:503:rajveer:/home/rajveer:/bin/bash
testrpm:x:504:504:testrpm:/home/testrpm:/bin/bash
[root@rdso ~]#
```

The above output shows that the home directories of these users are /home/vbg1, /home/websms, /home/rajveer and /home/testrpm, respectively. Ideally, we should change these locations because /home is also the folder where the home folders of local users are stored, and there might be situations in which you would want /home to remain intact. Therefore, change the homeDirectory attribute in your LDAP server. To do so, create a small ldif file as shown below and run the ldapmodify command.

```
[root@vbg ~]# cat /tmp/modify.ldif
dn: uid=vbg1,ou=People,dc=knafl,dc=org
changetype: modify
replace: homeDirectory
homeDirectory: /home/userhome/vbg1

dn: uid=websms,ou=People,dc=knafl,dc=org
changetype: modify
replace: homeDirectory
homeDirectory: /home/userhome/websms

dn: uid=rajveer,ou=People,dc=knafl,dc=org
changetype: modify
replace: homeDirectory
homeDirectory: /home/userhome/rajveer

dn: uid=testrpm,ou=People,dc=knafl,dc=org
changetype: modify
replace: homeDirectory
homeDirectory: /home/userhome/testrpm

[root@vbg ~]#
```

The above ldif file is self-explanatory. The various permissible values for changetype are: add, modify, delete, and modrdn. The 'modrdn' option is used when the relative distinguished name of an entry is about to change. We will look into these kinds of ldif files a little later in the series.

As of now, the ldif file specifies which DNs to modify and the values of which attributes to change. Let us run the ldapmodify command on the LDAP server =>

```
[root@vbg ~]# ldapmodify -xh localhost -D
"cn=Manager,dc=knafl,dc=org" -W -f /tmp/modify.ldif
Enter LDAP password:

modifying entry "uid=vbg1,ou=People,dc=knafl,dc=org"
modifying entry "uid=websms,ou=People,dc=knafl,dc=org"
modifying entry "uid=rajveer,ou=People,dc=knafl,dc=org"
modifying entry "uid=testrpm,ou=People,dc=knafl,dc=org"

[root@vbg ~]#
```

Before proceeding further, let us check from the client machine whether the changes have been successfully made:

```
[root@rdso ~]# getent passwd | tail -n 5
[root@rdso ~]# getent passwd | tail -n 5
```